

Appendix for “Code-Line-Level Defect Prediction with Confidence Enhancement and Counterfactual Analysis”

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1 The Results of CLEAR

This section provides a presentation and analysis of the experimental results reported in the main body of the paper. Specifically, we present the more complete results for the evaluated metrics and further examine the performance of CLEAR across different projects and baseline methods.

1.1 Results on the Primary Dataset

(1) Comparison of ranking performance.

The results further illustrate the performance characteristics of CLEAR across the nine studied projects. As shown in the Table 1 for the ranking-oriented metrics, CLEAR achieves the best Recall@Top20%LOC on eight of the nine projects. CLEAR-MAX performs best on ActiveMQ, Derby, HBase, and Wicket, while CLEAR-SUM, CLEAR-TOP2AVG, and CLEAR-TOP3AVG achieve the best results on two, one, and one projects, respectively. The only exception is Hive, where SOUND-OP2 and SOUND-TARANTULA achieve a higher Recall@Top20%LOC of 0.635 than the CLEAR variants. For Effort@Top20%Recall, CLEAR obtains the best result on five projects, whereas GLANCE-EA, GLANCE-MD, and SOUND-TARANTULA perform better on the remaining projects. A similar pattern is observed for IFA: CLEAR achieves the best result on six projects, with CLEAR-MAX performing best on three projects, CLEAR-SUM on two, and CLEAR-TOP2AVG on one. However, N-gram, LineDP, and SOUND-TARANTULA obtain lower IFA values on Camel, HBase, and Lucene, respectively. These results indicate that CLEAR generally provides favorable defective-line prioritization.

(2) Comparison of prediction accuracy.

As shown in Table 2, CLEAR achieves the best AUC on five of the nine projects. CLEAR-MAX obtains the best AUC on ActiveMQ, Derby, and Wicket, while CLEAR-SUM and CLEAR-TOP2AVG perform best on Lucene and Camel, respectively. On the remaining projects, GLANCE-EA, SOUND-BARINEL, and SOUND-OP2 achieve better AUC values. For D2H, CLEAR achieves the best performance on eight projects, with Hive being the only exception, where SOUND-OP2 and SOUND-TARANTULA obtain the lowest value of 0.303. CLEAR also achieves the best FAR on seven projects, while GLANCE-EA performs best on JRuby and SOUND-OP2/SOUND-TARANTULA obtain the best result on Hive. Compared with the differences observed for AUC and D2H, the absolute differences in FAR are generally small, suggesting that the methods exhibit relatively similar false-alarm behavior under the adopted prediction setting.

Across the six metrics, no single CLEAR aggregation strategy consistently performs best on all projects. CLEAR-MAX achieves the largest number of best results for Recall@Top20%LOC, AUC, D2H, and FAR, whereas CLEAR-SUM shows competitive performance particularly for Effort@Top20%Recall and IFA. CLEAR-TOP2AVG and CLEAR-TOP3AVG also achieve the best results on several individual projects. Overall, the results support the favorable performance of CLEAR observed in the main analysis, while also showing that its effectiveness and the relative strengths of different aggregation strategies vary across projects and evaluation metrics.

Table 1. Average ranking performance of CLEAR and the baseline methods on the nine studied projects. Bold values indicate the best result for each project. The #Best/9 column reports the number of projects on which each method achieves the best performance.

Metric	Method	ActiveMQ	Camel	Derby	Groovy	HBase	Hive	JRuby	Lucene	Wicket	#Best/9
Recall@Top20%LOC (↑)	GLANCE-LR	0.503	0.545	0.413	0.568	0.094	0.235	0.132	0.387	0.365	0
	GLANCE-EA	0.430	0.464	0.231	0.422	0.413	0.068	0.199	0.115	0.264	0
	GLANCE-MD	0.399	0.498	0.298	0.277	0.054	0.116	0.081	0.278	0.190	0
	LineDP	0.501	0.545	0.413	0.561	0.094	0.235	0.132	0.387	0.365	0
	DeepLineDP	0.373	0.304	0.307	0.237	0.281	0.259	0.200	0.259	0.285	0
	N-gram	0.375	0.304	0.305	0.232	0.314	0.259	0.200	0.259	0.281	0
	ErrorProne	0.374	0.304	0.310	0.234	0.249	0.259	0.200	0.259	0.277	0
	SOUND-BARINEL	0.503	0.545	0.413	0.568	0.094	0.235	0.132	0.387	0.365	0
	SOUND-DSTAR	0.503	0.544	0.413	0.568	0.094	0.235	0.132	0.387	0.365	0
	SOUND-OCHIAI	0.503	0.545	0.413	0.568	0.094	0.235	0.132	0.387	0.365	0
	SOUND-OP2	0.503	0.545	0.409	0.546	0.098	0.635	0.132	0.387	0.310	1
	SOUND-TARANTULA	0.502	0.545	0.409	0.546	0.098	0.635	0.132	0.387	0.310	1
	CLEAR-MAX	0.517	0.540	0.539	0.590	0.480	0.469	0.213	0.416	0.582	4
	CLEAR-SUM	0.498	0.581	0.518	0.592	0.479	0.479	0.192	0.471	0.578	2
	CLEAR-TOP2AVG	0.515	0.529	0.528	0.600	0.479	0.471	0.220	0.422	0.568	1
	CLEAR-TOP3AVG	0.511	0.538	0.530	0.589	0.479	0.447	0.232	0.427	0.576	1
Effort@Top20%Recall (↓)	GLANCE-LR	0.048	0.066	0.088	0.058	0.429	0.263	0.313	0.085	0.184	0
	GLANCE-EA	0.084	0.028	0.193	0.143	0.179	0.413	0.226	0.199	0.192	2
	GLANCE-MD	0.104	0.027	0.124	0.181	0.256	0.389	0.317	0.185	0.193	1
	LineDP	0.049	0.066	0.088	0.059	0.430	0.263	0.315	0.085	0.184	0
	DeepLineDP	0.115	0.093	0.140	0.219	0.429	0.261	0.410	0.111	0.136	0
	N-gram	0.115	0.093	0.137	0.218	0.428	0.261	0.410	0.110	0.137	0
	ErrorProne	0.116	0.092	0.139	0.219	0.430	0.261	0.410	0.111	0.137	0
	SOUND-BARINEL	0.048	0.066	0.088	0.056	0.430	0.263	0.314	0.085	0.184	0
	SOUND-DSTAR	0.048	0.066	0.088	0.057	0.430	0.263	0.315	0.085	0.184	0
	SOUND-OCHIAI	0.048	0.066	0.088	0.056	0.430	0.263	0.315	0.085	0.184	0
	SOUND-OP2	0.065	0.070	0.092	0.077	0.430	0.082	0.332	0.077	0.156	0
	SOUND-TARANTULA	0.065	0.070	0.092	0.076	0.430	0.082	0.331	0.077	0.156	1
	CLEAR-MAX	0.034	0.046	0.041	0.045	0.225	0.028	0.295	0.124	0.103	0
	CLEAR-SUM	0.033	0.044	0.041	0.034	0.224	0.028	0.284	0.099	0.099	3
	CLEAR-TOP2AVG	0.032	0.045	0.041	0.035	0.220	0.028	0.286	0.104	0.103	1
	CLEAR-TOP3AVG	0.033	0.046	0.041	0.025	0.217	0.029	0.284	0.106	0.103	1
IFA (↓)	GLANCE-LR	846	1659	7037	428	145	315	12991	2952	528	0
	GLANCE-EA	3962	4876	7794	449	413	82091	9777	43487	3057	0
	GLANCE-MD	5668	1041	9944	2578	496	581	16606	40315	847	0
	LineDP	888	1759	7525	195	139	371	13005	2946	1343	1
	DeepLineDP	1997	254	868	415	50172	35160	2718	4663	4546	0
	N-gram	1982	72	606	444	50159	35098	2196	4695	4543	1
	ErrorProne	2020	134	1175	422	50201	35252	2949	4696	4612	0
	SOUND-BARINEL	840	1652	6808	100	161	270	13254	2948	611	0
	SOUND-DSTAR	798	1624	6709	204	182	259	12871	2935	434	0
	SOUND-OCHIAI	816	1638	6751	105	166	227	13117	2937	464	0
	SOUND-OP2	545	352	77	2939	1853	310	2624	510	1343	0
	SOUND-TARANTULA	441	364	75	2909	1927	310	2598	505	1352	1
	CLEAR-MAX	147	337	7	4	1144	0	389	1249	404	3
	CLEAR-SUM	108	446	18	3	592	5	505	1039	454	2
	CLEAR-TOP2AVG	118	385	5	4	682	5	851	1455	473	1
	CLEAR-TOP3AVG	123	412	7	4	458	5	688	1552	547	0

Table 2. Average prediction accuracy of CLEAR and the baseline methods on the nine studied projects. Bold values indicate the best result for each project. The #Best/9 column reports the number of projects on which each method achieves the best performance.

Metric	Method	ActiveMQ	Camel	Derby	Groovy	HBase	Hive	JUnit	Lucene	Wicket	#Best/9
AUC (↑)	GLANCE-LR	0.694	0.688	0.680	0.782	0.571	0.573	0.493	0.549	0.632	0
	GLANCE-EA	0.676	0.717	0.545	0.677	0.731	0.408	0.555	0.426	0.546	2
	GLANCE-MD	0.631	0.679	0.612	0.665	0.686	0.439	0.539	0.503	0.562	0
	LineDP	0.694	0.688	0.680	0.781	0.568	0.572	0.492	0.549	0.631	0
	DeepLineDP	0.684	0.618	0.588	0.659	0.573	0.593	0.434	0.620	0.649	0
	N-gram	0.684	0.619	0.588	0.658	0.575	0.595	0.435	0.620	0.649	0
	ErrorProne	0.684	0.619	0.588	0.658	0.571	0.594	0.434	0.620	0.648	0
	SOUND-BARINEL	0.694	0.688	0.680	0.782	0.572	0.572	0.492	0.549	0.632	1
	SOUND-DSTAR	0.694	0.688	0.680	0.782	0.570	0.572	0.492	0.549	0.632	0
	SOUND-OCHIAI	0.694	0.688	0.680	0.782	0.571	0.572	0.492	0.549	0.632	0
	SOUND-OP2	0.691	0.687	0.679	0.761	0.575	0.723	0.497	0.548	0.635	1
	SOUND-TARANTULA	0.691	0.687	0.679	0.762	0.575	0.723	0.497	0.548	0.635	0
	CLEAR-MAX	0.734	0.763	0.724	0.773	0.640	0.690	0.485	0.673	0.777	3
	CLEAR-SUM	0.723	0.753	0.715	0.774	0.649	0.692	0.489	0.703	0.764	1
D2H (↓)	CLEAR-TOP2AVG	0.733	0.766	0.718	0.776	0.641	0.691	0.485	0.675	0.777	1
	CLEAR-TOP3AVG	0.733	0.754	0.716	0.772	0.650	0.684	0.487	0.677	0.774	0
	GLANCE-LR	0.379	0.352	0.439	0.338	0.656	0.560	0.630	0.460	0.473	0
	GLANCE-EA	0.427	0.405	0.562	0.433	0.453	0.674	0.584	0.642	0.540	0
	GLANCE-MD	0.448	0.383	0.516	0.531	0.684	0.641	0.665	0.531	0.590	0
	LineDP	0.381	0.353	0.439	0.343	0.656	0.560	0.630	0.460	0.473	0
	DeepLineDP	0.466	0.512	0.510	0.558	0.530	0.545	0.584	0.543	0.525	0
	N-gram	0.465	0.512	0.511	0.561	0.509	0.545	0.584	0.543	0.528	0
	ErrorProne	0.465	0.512	0.508	0.560	0.551	0.545	0.584	0.543	0.531	0
	SOUND-BARINEL	0.379	0.353	0.439	0.338	0.656	0.560	0.630	0.460	0.473	0
	SOUND-DSTAR	0.379	0.353	0.439	0.338	0.656	0.560	0.630	0.460	0.473	0
	SOUND-OCHIAI	0.379	0.353	0.439	0.338	0.656	0.560	0.630	0.460	0.473	0
	SOUND-OP2	0.379	0.353	0.441	0.353	0.654	0.303	0.630	0.460	0.509	1
	SOUND-TARANTULA	0.380	0.353	0.441	0.353	0.654	0.303	0.630	0.460	0.509	1
FAR (↓)	CLEAR-MAX	0.370	0.355	0.355	0.323	0.421	0.401	0.576	0.441	0.345	4
	CLEAR-SUM	0.383	0.329	0.369	0.322	0.421	0.394	0.589	0.404	0.347	2
	CLEAR-TOP2AVG	0.371	0.362	0.362	0.318	0.421	0.400	0.571	0.436	0.354	1
	CLEAR-TOP3AVG	0.374	0.356	0.361	0.324	0.421	0.416	0.562	0.433	0.349	1
	GLANCE-LR	0.19929	0.19967	0.19950	0.19914	0.20094	0.19983	0.20122	0.19942	0.19959	0
	GLANCE-EA	0.19948	0.19975	0.19993	0.19951	0.19806	0.20028	0.19977	0.20011	0.19980	1
	GLANCE-MD	0.19951	0.19972	0.19971	0.19987	0.20130	0.20016	0.20128	0.19964	0.20002	0
	LineDP	0.19930	0.19967	0.19950	0.19915	0.20094	0.19983	0.20122	0.19942	0.19959	0
	DeepLineDP	0.19955	0.19978	0.19960	0.19993	0.19878	0.19958	0.20116	0.19979	0.19976	0
	N-gram	0.19954	0.19978	0.19960	0.19994	0.19829	0.19958	0.20116	0.19979	0.19976	0
	ErrorProne	0.19955	0.19978	0.19959	0.19993	0.19923	0.19958	0.20116	0.19979	0.19977	0
	SOUND-BARINEL	0.19929	0.19967	0.19950	0.19914	0.20094	0.19983	0.20122	0.19942	0.19959	0
	SOUND-DSTAR	0.19929	0.19967	0.19950	0.19914	0.20094	0.19983	0.20122	0.19942	0.19959	0
	SOUND-OCHIAI	0.19929	0.19967	0.19950	0.19914	0.20094	0.19983	0.20122	0.19942	0.19959	0
	SOUND-OP2	0.19929	0.19967	0.19942	0.19918	0.20091	0.19907	0.20122	0.19942	0.19972	1
	SOUND-TARANTULA	0.19929	0.19967	0.19942	0.19918	0.20091	0.19907	0.20122	0.19942	0.19972	1
	CLEAR-MAX	0.19907	0.19962	0.19864	0.19882	0.19707	0.19926	0.20114	0.19933	0.19905	4
	CLEAR-SUM	0.19913	0.19958	0.19875	0.19881	0.19707	0.19924	0.20113	0.19928	0.19906	2
	CLEAR-TOP2AVG	0.19908	0.19962	0.19873	0.19879	0.19707	0.19926	0.20111	0.19932	0.19907	1
	CLEAR-TOP3AVG	0.19908	0.19961	0.19872	0.19882	0.19707	0.19932	0.20112	0.19933	0.19905	0

1.2 Results on the Additional Dataset

(1) Comparison of ranking performance.

As shown in Table 3, CLEAR generally achieves favorable ranking performance on the additional dataset. For Recall@Top20%LOC, CLEAR-TOP2AVG and CLEAR-TOP3AVG exhibit strong overall performance among the compared methods. At the project level, CLEAR-TOP3AVG obtains the

best result on 8 of the 19 projects, while CLEAR-TOP2AVG does so on 5 projects, indicating that CLEAR can identify a larger proportion of defective lines within a limited inspection budget on many projects. For Effort@Top20%Recall, CLEAR variants also require less inspection effort than most baselines. In particular, CLEAR-TOP2AVG and CLEAR-TOP3AVG each achieve the best result on six projects, suggesting that CLEAR can often locate 20% of the defective lines after inspecting a relatively small portion of the code. For IFA, CLEAR variants generally place the first defective line earlier in the ranking, thereby reducing the number of clean lines that developers need to inspect before encountering a defect. CLEAR-TOP3AVG achieves the best IFA on eight projects, while CLEAR-SUM does so on six projects. However, no single aggregation variant dominates all ranking-oriented metrics or all projects. Overall, the results suggest that CLEAR maintains strong ranking effectiveness on the additional dataset, while the relative advantage of different aggregation strategies depends on the specific project and evaluation metric.

(2) Comparison of prediction accuracy.

As shown in Table 4, CLEAR also achieves competitive prediction accuracy on the additional dataset. For AUC, CLEAR-MAX and CLEAR-TOP3AVG achieve strong performance across the studied projects, and CLEAR-MAX obtains the best result on 8 of the 19 projects, while CLEAR-TOP3AVG does so on 6 projects. This indicates that CLEAR generally provides better discrimination between defective and clean lines in the overall ranking. For D2H, CLEAR-TOP3AVG achieves the best result on 8 projects, while CLEAR-MAX and CLEAR-TOP2AVG each do so on 4 projects, indicating a favorable overall balance between recall and false alarm rate. For FAR, the absolute differences among most methods are relatively small, with the values concentrated within a narrow range. Nevertheless, CLEAR-TOP3AVG achieves the lowest FAR on 8 projects, while CLEAR-MAX and CLEAR-TOP2AVG each achieve the best result on two projects. These results show that different CLEAR variants exhibit complementary strengths across the accuracy-oriented metrics. CLEAR maintains favorable accuracy-oriented performance on the additional dataset, while different aggregation variants exhibit complementary strengths across AUC, D2H, and FAR.

Overall, the results on the additional dataset are broadly consistent with those observed in the primary study. CLEAR performs favorably across both ranking-oriented and accuracy-oriented metrics. In particular, CLEAR-TOP2AVG and CLEAR-TOP3AVG show strong performance in Recall@Top20%LOC and Effort@Top20%Recall, CLEAR-MAX and CLEAR-TOP3AVG perform favorably in AUC, while CLEAR-TOP3AVG also shows strong performance in D2H and FAR. Although the best-performing aggregation variant differs across metrics and projects, the overall results indicate that the effectiveness of CLEAR is not limited to the primary dataset.

2 Statistical Tests

To assess whether the observed performance differences are consistent across releases, we further conduct statistical tests to examine their statistical significance and practical magnitude. We therefore conduct statistical tests to determine whether these differences are consistent enough to be considered statistically significant and to assess their practical magnitude.

We follow the statistical testing procedures adopted in GLANCE [10] and SOUND [30]. Specifically, we employ the Wilcoxon signed-rank test and Cliff's Delta to compare CLEAR with the baseline methods using paired release results across the studied projects. The Wilcoxon signed-rank test is used to examine whether the performance differences are statistically significant, while Cliff's Delta quantifies the magnitude of the differences. To account for multiple comparisons, we adjust the resulting p -values using the Benjamini-Hochberg (BH) procedure.

Based on the overall performance across the two datasets and six evaluation metrics, we select representative configurations for the statistical comparison. Specifically, CLEAR-SUM, SOUND-OP2, and GLANCE-LR are selected to represent CLEAR, SOUND, and GLANCE, respectively.

Table 3. Comparison of ranking performance on the additional dataset. Due to space limitations, project names are abbreviated using their first three letters (e.g., Amb denotes Ambari). Each value represents the average performance of a method on the corresponding project, and the best result(s) are highlighted in **bold**.

Metric	Method	Amb	Act	Boo	Cal	Cas	Fli	Gro	HBa	Hiv	Ign	Log	Mah	Mav	NiF	Nut	Sto	Tik	Str	Zoo	#Best/19
Recall@Top20%LOC (↑)	GLANCE-LR	0.366	0.469	0.452	0.291	0.276	0.128	0.464	0.359	0.474	0.241	0.349	0.347	0.582	0.387	0.424	0.721	0.288	0.677	0.452	0
	GLANCE-EA	0.330	0.343	0.341	0.195	0.016	0.136	0.279	0.070	0.125	0.259	0.192	0.063	0.135	0.270	0.258	0.119	0.087	0.306	0.316	0
	GLANCE-MD	0.243	0.337	0.376	0.185	0.040	0.091	0.167	0.165	0.215	0.150	0.119	0.080	0.093	0.232	0.236	0.499	0.062	0.310	0.276	0
	LineDP	0.365	0.468	0.451	0.291	0.276	0.128	0.463	0.359	0.474	0.245	0.350	0.347	0.581	0.387	0.421	0.721	0.290	0.676	0.450	0
	DeepLineDP	0.421	0.542	0.173	0.381	0.483	0.569	0.541	0.367	0.513	0.331	0.381	0.531	0.484	0.292	0.362	0.762	0.420	0.459	0.178	3
	N-gram	0.422	0.542	0.172	0.381	0.483	0.569	0.541	0.368	0.513	0.330	0.382	0.531	0.480	0.291	0.363	0.762	0.420	0.460	0.179	1
	ErrorProne	0.421	0.543	0.167	0.381	0.483	0.569	0.540	0.342	0.513	0.330	0.381	0.531	0.481	0.290	0.357	0.762	0.419	0.460	0.179	1
	SOUND-BARINEL	0.366	0.470	0.452	0.291	0.276	0.128	0.464	0.361	0.474	0.246	0.351	0.347	0.590	0.387	0.428	0.721	0.293	0.678	0.453	1
	SOUND-DSTAR	0.366	0.468	0.448	0.291	0.276	0.128	0.464	0.360	0.474	0.243	0.349	0.347	0.581	0.387	0.426	0.721	0.291	0.677	0.452	0
	SOUND-OCHIAI	0.366	0.469	0.449	0.291	0.276	0.128	0.464	0.361	0.474	0.246	0.350	0.347	0.587	0.387	0.428	0.721	0.293	0.677	0.453	0
	SOUND-OP2	0.305	0.529	0.313	0.266	0.502	0.128	0.411	0.407	0.474	0.210	0.411	0.433	0.666	0.418	0.530	0.721	0.352	0.707	0.380	1
	SOUND-TARANTULA	0.305	0.530	0.313	0.266	0.502	0.128	0.415	0.408	0.474	0.210	0.411	0.433	0.667	0.418	0.533	0.721	0.352	0.707	0.380	0
	CLEAR-MAX	0.528	0.694	0.372	0.653	0.703	0.314	0.503	0.740	0.472	0.604	0.654	0.630	0.733	0.655	0.684	0.827	0.582	0.699	0.590	2
	CLEAR-SUM	0.516	0.689	0.408	0.666	0.702	0.384	0.496	0.772	0.422	0.582	0.646	0.627	0.732	0.634	0.674	0.776	0.581	0.686	0.478	0
	CLEAR-TOP2AVG	0.535	0.693	0.398	0.694	0.691	0.338	0.498	0.703	0.474	0.628	0.650	0.634	0.733	0.651	0.688	0.828	0.580	0.690	0.509	5
	CLEAR-TOP3AVG	0.532	0.694	0.402	0.720	0.715	0.413	0.501	0.687	0.428	0.638	0.657	0.636	0.733	0.656	0.686	0.763	0.584	0.688	0.482	8
Effort@Top20%Recall (↓)	GLANCE-LR	0.117	0.087	0.112	0.109	0.234	0.451	0.119	0.116	0.089	0.174	0.139	0.158	0.097	0.083	0.116	0.016	0.182	0.068	0.101	0
	GLANCE-EA	0.122	0.128	0.093	0.192	0.452	0.282	0.157	0.400	0.244	0.185	0.220	0.286	0.259	0.169	0.163	0.411	0.308	0.115	0.154	0
	GLANCE-MD	0.151	0.150	0.163	0.203	0.546	0.380	0.231	0.424	0.191	0.248	0.281	0.400	0.376	0.163	0.198	0.071	0.396	0.167	0.235	0
	LineDP	0.118	0.087	0.112	0.110	0.234	0.451	0.120	0.116	0.088	0.175	0.139	0.158	0.097	0.083	0.117	0.016	0.182	0.068	0.101	0
	DeepLineDP	0.106	0.057	0.241	0.104	0.104	0.072	0.081	0.162	0.052	0.128	0.133	0.056	0.069	0.149	0.096	0.047	0.107	0.101	0.216	1
	N-gram	0.106	0.057	0.241	0.104	0.104	0.071	0.081	0.146	0.052	0.127	0.133	0.057	0.068	0.149	0.096	0.047	0.107	0.101	0.217	0
	ErrorProne	0.106	0.058	0.242	0.103	0.104	0.072	0.082	0.167	0.052	0.128	0.133	0.056	0.070	0.149	0.097	0.046	0.107	0.101	0.216	0
	SOUND-BARINEL	0.117	0.086	0.113	0.109	0.234	0.451	0.118	0.116	0.088	0.173	0.138	0.158	0.097	0.083	0.116	0.016	0.182	0.068	0.101	0
	SOUND-DSTAR	0.118	0.087	0.113	0.110	0.234	0.451	0.119	0.116	0.089	0.174	0.139	0.158	0.097	0.083	0.116	0.016	0.182	0.068	0.101	0
	SOUND-OCHIAI	0.117	0.087	0.113	0.109	0.234	0.451	0.119	0.116	0.088	0.174	0.138	0.158	0.097	0.083	0.116	0.016	0.182	0.068	0.101	0
	SOUND-OP2	0.163	0.086	0.128	0.145	0.076	0.451	0.125	0.115	0.083	0.190	0.099	0.105	0.028	0.083	0.061	0.039	0.122	0.046	0.131	0
	SOUND-TARANTULA	0.163	0.086	0.128	0.145	0.076	0.451	0.124	0.115	0.083	0.190	0.099	0.105	0.028	0.083	0.060	0.039	0.122	0.046	0.128	0
	CLEAR-MAX	0.077	0.023	0.116	0.020	0.012	0.115	0.064	0.018	0.076	0.016	0.035	0.016	0.014	0.013	0.025	0.007	0.049	0.023	0.065	3
	CLEAR-SUM	0.081	0.024	0.078	0.021	0.015	0.114	0.064	0.022	0.076	0.026	0.035	0.018	0.014	0.020	0.026	0.008	0.049	0.022	0.063	2
	CLEAR-TOP2AVG	0.076	0.023	0.111	0.017	0.013	0.108	0.063	0.019	0.075	0.016	0.033	0.016	0.015	0.013	0.026	0.006	0.048	0.022	0.066	6
	CLEAR-TOP3AVG	0.077	0.022	0.090	0.017	0.012	0.108	0.064	0.020	0.077	0.016	0.033	0.016	0.015	0.013	0.026	0.007	0.048	0.022	0.071	6
IFA (↓)	GLANCE-LR	431	142	671	247	7351	2	874	7	3511	582	4	117	198	0	96	735	95	100	156	2
	GLANCE-EA	961	1179	20	990	28184	502	1866	370	2062	116	4	960	1541	544	433	2350	327	957	3495	1
	GLANCE-MD	1199	1564	658	776	11304	567	3909	366	3974	577	53	475	482	1577	315	5916	60	1943	6076	0
	LineDP	436	158	662	281	7371	33	934	43	3586	574	6	106	210	0	102	725	152	71	159	1
	DeepLineDP	173	130	372	8	174	888	577	13741	343	329	127	836	206	392	110	746	72	109	1266	0
	N-gram	201	162	160	22	171	901	535	13045	331	326	84	833	42	141	72	718	62	115	1426	0
	ErrorProne	224	158	204	34	232	931	596	13310	304	432	105	828	325	553	169	760	96	166	1431	0
	SOUND-BARINEL	424	131	684	249	7348	8	710	3	3389	579	0	89	208	2	83	728	38	65	135	1
	SOUND-DSTAR	437	138	668	294	7384	7	947	49	3473	564	15	137	229	2	128	739	77	70	154	0
	SOUND-OCHIAI	432	133	684	268	7371	18	845	29	3523	560	10	110	209	2	101	734	69	66	144	0
	SOUND-OP2	92	225	103	11	41	802	589	96	916	137	81	60	16	0	73	625	34	14	101	1
	SOUND-TARANTULA	84	193	151	0	37	795	550	92	883	136	67	51	14	0	53	554	24	12	73	3
	CLEAR-MAX	67	5	30	0	32	2125	70	2	194	2	1	7	16	0	61	7	12	12	245	2
	CLEAR-SUM	42	1	43	1	12	2231	86	5	6	0	0	5	13	18	60	7	0	7	252	6
	CLEAR-TOP2AVG	85	2	75	0	10	2212	68	5	201	0	2	4	14	11	59	7	12	8	224	3
	CLEAR-TOP3AVG	34	2	214	0	8	2209	70	2	190	0	2	3	13	10	59	9	11	6	184	8

Following prior work [10, 30], we further summarize the comparisons using Win/Tie/Loss (W/T/L). The magnitude of Cliff’s Delta is classified as negligible (N, $|\delta| < 0.147$), small (S, $0.147 \leq |\delta| < 0.33$), medium (M, $0.33 \leq |\delta| < 0.474$), or large (L, $|\delta| \geq 0.474$). For Recall@Top20%LOC and AUC, where larger values are better, a positive δ favors CLEAR. In contrast, for Effort@Top20%Recall, IFA, D2H, and FAR, where smaller values are better, a negative δ favors CLEAR. Based on these criteria, a comparison is regarded as a Win (W) when the BH-corrected p -value is smaller than 0.05 and Cliff’s Delta indicates a non-negligible effect ($|\delta| \geq 0.147$) in favor of CLEAR. A statistically significant and non-negligible effect in the opposite direction is regarded as a Loss (L); otherwise, the comparison is regarded as a Tie (T).

Table 5 presents the statistical test results, while Table 6 summarizes the corresponding W/T/L outcomes. On the primary dataset, CLEAR-SUM achieves 34 Wins, two Ties, and no Losses across the 36 pairwise comparisons. The two Ties occur against SOUND-OP2 for Recall@Top20%LOC and IFA, while CLEAR-SUM achieves 6/0/0 against the baselines for Effort@Top20%Recall, AUC, D2H, and FAR. The significant comparisons are generally associated with small-to-large effect sizes, with particularly strong effects observed against DeepLineDP, N-gram, and ErrorProne for several metrics. The additional dataset provides even stronger statistical evidence. CLEAR-SUM achieves 6/0/0 for each of the six metrics, corresponding to 36 Wins with no Ties or Losses. The effect sizes are consistently large for Recall@Top20%LOC, Effort@Top20%Recall, IFA, and D2H, range from medium to large for AUC, and from small to medium for FAR. These results indicate that the

Table 4. Comparison of prediction accuracy on the additional dataset. Due to space limitations, project names are abbreviated using their first three letters (e.g., Amb denotes Ambari). Each value represents the average performance of a method on the corresponding project, and the best result(s) are highlighted in **bold**.

Metric	Method	Amb	Act	Boo	Cal	Cas	Flt	Gro	HBs	Hrv	Ign	Log	Mah	Mav	Nif	Nut	Sto	Tik	Str	Zoo	#Best/19
AUC (↑)	GLANCE-LR	0.686	0.773	0.680	0.667	0.627	0.426	0.703	0.667	0.626	0.645	0.691	0.703	0.805	0.689	0.758	0.772	0.680	0.787	0.695	0
	GLANCE-EA	0.605	0.633	0.514	0.507	0.341	0.565	0.586	0.372	0.476	0.588	0.521	0.490	0.521	0.543	0.509	0.455	0.493	0.572	0.560	0
	GLANCE-MD	0.569	0.579	0.622	0.537	0.344	0.471	0.547	0.456	0.493	0.534	0.463	0.422	0.460	0.520	0.517	0.538	0.427	0.581	0.534	0
	LineDP	0.686	0.773	0.679	0.667	0.626	0.426	0.702	0.671	0.625	0.644	0.690	0.703	0.804	0.689	0.757	0.771	0.679	0.787	0.693	0
	DeepLineDP	0.699	0.794	0.513	0.685	0.748	0.764	0.746	0.704	0.702	0.671	0.672	0.731	0.769	0.640	0.649	0.827	0.706	0.724	0.592	0
	N-gran	0.699	0.794	0.514	0.685	0.749	0.764	0.746	0.709	0.702	0.672	0.672	0.731	0.769	0.640	0.649	0.827	0.706	0.724	0.592	2
	ErrorProne	0.699	0.794	0.513	0.684	0.747	0.764	0.746	0.701	0.701	0.671	0.671	0.731	0.767	0.640	0.649	0.827	0.706	0.724	0.590	0
	SOUND-BARBEL	0.687	0.773	0.679	0.668	0.628	0.426	0.704	0.683	0.626	0.645	0.691	0.703	0.807	0.689	0.750	0.773	0.681	0.787	0.696	0
	SOUND-DSTAR	0.686	0.773	0.679	0.667	0.627	0.426	0.703	0.680	0.626	0.644	0.691	0.703	0.806	0.689	0.758	0.772	0.680	0.787	0.695	0
	SOUND-OCCHAI	0.686	0.773	0.679	0.667	0.627	0.426	0.703	0.681	0.626	0.645	0.691	0.703	0.807	0.689	0.759	0.772	0.681	0.787	0.695	0
D2H (↓)	SOUND-ORP	0.667	0.782	0.653	0.626	0.745	0.426	0.697	0.676	0.626	0.615	0.738	0.727	0.838	0.704	0.834	0.771	0.714	0.804	0.654	0
	SOUND-TARANTULA	0.667	0.782	0.653	0.626	0.745	0.426	0.698	0.679	0.626	0.615	0.739	0.727	0.839	0.705	0.785	0.771	0.715	0.804	0.655	0
	CLEAR-MAX	0.743	0.848	0.685	0.823	0.839	0.591	0.759	0.839	0.690	0.780	0.816	0.817	0.865	0.788	0.834	0.900	0.768	0.837	0.710	8
	CLEAR-SUM	0.733	0.845	0.704	0.822	0.831	0.592	0.756	0.825	0.677	0.772	0.814	0.813	0.856	0.778	0.828	0.889	0.768	0.834	0.703	1
	CLEAR-TOP20%G	0.744	0.848	0.695	0.833	0.836	0.592	0.757	0.829	0.686	0.786	0.817	0.819	0.863	0.790	0.833	0.897	0.768	0.836	0.709	2
	CLEAR-TOP10%G	0.741	0.809	0.790	0.841	0.838	0.597	0.756	0.833	0.645	0.749	0.831	0.819	0.862	0.794	0.831	0.891	0.769	0.835	0.702	6
	GLANCE-LR	0.471	0.401	0.413	0.520	0.534	0.633	0.409	0.475	0.398	0.555	0.481	0.483	0.530	0.456	0.430	0.261	0.524	0.284	0.415	0
	GLANCE-EA	0.494	0.485	0.487	0.587	0.710	0.627	0.529	0.673	0.635	0.542	0.588	0.478	0.628	0.535	0.543	0.639	0.661	0.510	0.504	0
	GLANCE-MD	0.553	0.489	0.465	0.594	0.694	0.658	0.606	0.607	0.573	0.618	0.639	0.666	0.657	0.561	0.558	0.383	0.679	0.509	0.531	0
	LineDP	0.471	0.402	0.413	0.520	0.534	0.633	0.409	0.475	0.398	0.552	0.480	0.483	0.531	0.456	0.432	0.261	0.522	0.285	0.416	0
FAR (↓)	DeepLineDP	0.434	0.352	0.602	0.458	0.392	0.335	0.358	0.469	0.373	0.493	0.461	0.359	0.392	0.520	0.473	0.244	0.433	0.408	0.598	1
	N-gran	0.433	0.352	0.603	0.457	0.392	0.335	0.358	0.468	0.373	0.494	0.460	0.359	0.394	0.521	0.472	0.244	0.434	0.408	0.598	1
	ErrorProne	0.433	0.352	0.606	0.458	0.391	0.335	0.358	0.487	0.373	0.494	0.460	0.359	0.394	0.521	0.476	0.244	0.434	0.407	0.598	1
	SOUND-BARBEL	0.471	0.400	0.413	0.520	0.534	0.633	0.408	0.473	0.398	0.551	0.479	0.483	0.525	0.456	0.428	0.261	0.520	0.284	0.414	1
	SOUND-DSTAR	0.471	0.402	0.415	0.520	0.534	0.633	0.409	0.474	0.398	0.553	0.481	0.483	0.531	0.456	0.429	0.261	0.522	0.284	0.415	0
	SOUND-OCCHAI	0.471	0.401	0.415	0.520	0.534	0.633	0.409	0.474	0.398	0.551	0.480	0.483	0.527	0.456	0.428	0.261	0.520	0.284	0.414	0
	SOUND-ORP	0.512	0.366	0.506	0.537	0.379	0.633	0.441	0.442	0.398	0.576	0.439	0.424	0.276	0.435	0.361	0.261	0.480	0.263	0.463	0
	SOUND-TARANTULA	0.512	0.366	0.506	0.537	0.379	0.633	0.439	0.442	0.398	0.576	0.439	0.424	0.276	0.435	0.358	0.261	0.480	0.263	0.463	0
	CLEAR-MAX	0.370	0.263	0.465	0.276	0.255	0.505	0.388	0.233	0.399	0.311	0.281	0.297	0.240	0.282	0.266	0.206	0.325	0.259	0.383	4
	CLEAR-SUM	0.378	0.266	0.441	0.268	0.255	0.458	0.393	0.270	0.434	0.325	0.286	0.299	0.241	0.295	0.272	0.241	0.326	0.266	0.396	0
CLEAR-TOP20%G	0.365	0.264	0.448	0.259	0.262	0.489	0.392	0.256	0.398	0.296	0.283	0.294	0.240	0.285	0.244	0.288	0.326	0.263	0.378	4	
FAR (↓)	CLEAR-TOP10%G	0.367	0.264	0.446	0.234	0.247	0.438	0.390	0.262	0.429	0.290	0.280	0.293	0.240	0.281	0.265	0.250	0.324	0.265	0.393	8
	GLANCE-LR	0.19741	0.19616	0.19728	0.19475	0.19947	0.20015	0.19821	0.19822	0.19922	0.19900	0.19580	0.19773	0.19758	0.19769	0.19665	0.19868	0.19759	0.19428	0.19520	1
	GLANCE-EA	0.19717	0.19746	0.19845	0.20110	0.20242	0.20013	0.19939	0.20138	0.20054	0.19855	0.19993	0.20266	0.20038	0.19903	0.19925	0.19988	0.20369	0.19911	0.19648	0
	GLANCE-MD	0.19819	0.19755	0.19837	0.20157	0.20198	0.20022	0.20018	0.20045	0.20011	0.20128	0.20206	0.20250	0.20074	0.19962	0.19962	0.19881	0.20466	0.20039	0.19698	0
	LineDP	0.19743	0.19617	0.19729	0.19947	0.19947	0.20015	0.19822	0.19822	0.19922	0.19891	0.19576	0.19773	0.19759	0.19770	0.19670	0.19868	0.19756	0.19431	0.19525	0
	DeepLineDP	0.19574	0.19567	0.20031	0.19854	0.19509	0.19916	0.19713	0.19715	0.19841	0.19619	0.19606	0.19263	0.19757	0.19869	0.19728	0.19829	0.19265	0.19640	0.20065	4
	N-gran	0.19572	0.19368	0.20034	0.19854	0.19508	0.19916	0.19713	0.19713	0.19841	0.19619	0.19604	0.19263	0.19763	0.19870	0.19726	0.19829	0.19266	0.19640	0.20065	3
	ErrorProne	0.19573	0.19367	0.20041	0.19855	0.19508	0.19916	0.19714	0.19759	0.19841	0.19619	0.19605	0.19262	0.19762	0.19873	0.19734	0.19829	0.19271	0.19639	0.20065	2
	SOUND-BARBEL	0.19741	0.19614	0.19729	0.19974	0.19947	0.20015	0.19821	0.19818	0.19922	0.19888	0.19571	0.19772	0.19752	0.19769	0.19659	0.19688	0.19749	0.19427	0.19518	0
	SOUND-DSTAR	0.19741	0.19617	0.19732	0.19945	0.19947	0.20015	0.19821	0.19821	0.19922	0.19895	0.19579	0.19773	0.19758	0.19769	0.19662	0.19688	0.19753	0.19428	0.19520	0
SOUND-OCCHAI	0.19741	0.19615	0.19731	0.19945	0.19947	0.20015	0.19821	0.19819	0.19922	0.19888	0.19577	0.19773	0.19754	0.19769	0.19660	0.19668	0.19750	0.19428	0.19518	1	
SOUND-ORP	0.19908	0.19581	0.19853	0.19624	0.19612	0.20015	0.19873	0.19774	0.19922	0.19975	0.19467	0.19527	0.19685	0.19715	0.19532	0.19688	0.19556	0.19294	0.19294	0.19839	1
SOUND-TARANTULA	0.19908	0.19580	0.19853	0.19623	0.19612	0.20015	0.19872	0.19774	0.19922	0.19975	0.19466	0.19526	0.19685	0.19715	0.19528	0.19688	0.19556	0.19294	0.19839	0	
FAR (↓)	CLEAR-MAX	0.19668	0.19295	0.19845	0.19780	0.19662	0.19978	0.19810	0.19376	0.19900	0.18970	0.19078	0.19256	0.19674	0.19374	0.19352	0.19872	0.18699	0.19408	0.19630	2
	CLEAR-SUM	0.19684	0.19803	0.19818	0.19721	0.19663	0.19965	0.19815	0.19466	0.19907	0.19026	0.19093	0.19283	0.19675	0.19594	0.19367	0.19888	0.18721	0.19423	0.19631	0
	CLEAR-TOP20%G	0.19656	0.19297	0.19826	0.17578	0.19479	0.19974	0.19814	0.19433	0.19898	0.18910	0.19084	0.19248	0.19674	0.19580	0.19349	0.19873	0.18710	0.19417	0.19621	2
	CLEAR-TOP10%G	0.19603	0.19296	0.19823	0.17451	0.19453	0.19959	0.19812	0.19439	0.19905	0.18886	0.19069	0.19245	0.19674	0.19576	0.19350	0.19894	0.18687	0.19417	0.19636	8

performance improvements of CLEAR-SUM are statistically supported across both ranking-oriented and accuracy-oriented metrics.

The W/T/L results in Table 6 further summarize the consistency of these comparisons across metrics. On the primary dataset, CLEAR-SUM achieves Wins against five of the six baselines for both Recall@Top20%LOC and IFA, while the remaining comparison resulting in a Tie, while it achieves Wins against all six baselines for Effort@Top20%Recall, AUC, D2H, and FAR. On the additional dataset, CLEAR-SUM achieves 6/0/0 across all six metrics. These results show that the advantages of CLEAR-SUM are consistently observed across both datasets, with only limited ties on the primary dataset and no statistically supported losses.

Overall, the statistical tests consistently support the effectiveness of CLEAR-SUM across the two datasets. CLEAR-SUM achieves statistically significant and non-negligible improvements in the vast majority of comparisons, with no statistically supported Losses observed across the six evaluation metrics.

Summary for Statistical Tests: The statistical tests further support the effectiveness of CLEAR across both datasets. CLEAR-SUM achieves 34 Wins on the primary dataset and 36 Wins on the additional dataset, with no Losses and predominantly non-negligible effect sizes. These results provide statistical evidence for the effectiveness of CLEAR across both ranking-oriented and accuracy-oriented metrics.

Table 5. Statistical test results between CLEAR-SUM and the baselines on the primary and additional datasets.

Dataset	Metric	Statistic	SOUND-OP2	GLANCE-LR	LineDP	DeepLineDP	N-gram	ErrorProne
Primary dataset [42]	Recall@Top20%LOC (↑)	BH-corrected p	0.053	0.017	0.017	<0.001	<0.001	<0.001
		Cliff's δ	0.217 (S)	0.248 (S)	0.251 (S)	0.541 (L)	0.533 (L)	0.556 (L)
		Result	T	W	W	W	W	W
	Effort@Top20%Recall (↓)	BH-corrected p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
		Cliff's δ	-0.384 (M)	-0.369 (M)	-0.380 (M)	-0.561 (L)	-0.561 (L)	-0.565 (L)
		Result	W	W	W	W	W	W
	IFA (↓)	BH-corrected p	0.061	0.013	0.003	<0.001	<0.001	<0.001
		Cliff's δ	-0.338 (M)	-0.372 (M)	-0.408 (M)	-0.578 (L)	-0.535 (L)	-0.645 (L)
		Result	T	W	W	W	W	W
	AUC (↑)	BH-corrected p	0.020	0.009	0.009	<0.001	<0.001	<0.001
		Cliff's δ	0.267 (S)	0.323 (S)	0.323 (S)	0.444 (M)	0.437 (M)	0.444 (M)
		Result	W	W	W	W	W	W
	D2H (↓)	BH-corrected p	0.0498	0.017	0.017	<0.001	<0.001	<0.001
		Cliff's δ	-0.217 (S)	-0.248 (S)	-0.251 (S)	-0.539 (L)	-0.527 (L)	-0.554 (L)
		Result	W	W	W	W	W	W
Additional dataset [10]	Recall@Top20%LOC (↑)	BH-corrected p	0.007	0.001	0.001	<0.001	<0.001	<0.001
		Cliff's δ	-0.214 (S)	-0.259 (S)	-0.259 (S)	-0.350 (M)	-0.350 (M)	-0.346 (M)
		Result	W	W	W	W	W	W
	Effort@Top20%Recall (↓)	BH-corrected p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
		Cliff's δ	-0.712 (L)	-0.760 (L)	-0.762 (L)	-0.689 (L)	-0.689 (L)	-0.690 (L)
		Result	W	W	W	W	W	W
	IFA (↓)	BH-corrected p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
		Cliff's δ	-0.550 (L)	-0.613 (L)	-0.657 (L)	-0.732 (L)	-0.703 (L)	-0.848 (L)
		Result	W	W	W	W	W	W
	AUC (↑)	BH-corrected p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
		Cliff's δ	0.421 (M)	0.486 (L)	0.489 (L)	0.551 (L)	0.551 (L)	0.553 (L)
		Result	W	W	W	W	W	W
	D2H (↓)	BH-corrected p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
		Cliff's δ	-0.479 (L)	-0.548 (L)	-0.549 (L)	-0.562 (L)	-0.562 (L)	-0.564 (L)
		Result	W	W	W	W	W	W
	FAR (↓)	BH-corrected p	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
		Cliff's δ	-0.315 (S)	-0.396 (M)	-0.396 (M)	-0.242 (S)	-0.244 (S)	-0.248 (S)
		Result	W	W	W	W	W	W

Table 6. W/T/L results of CLEAR against the baselines on the primary and additional datasets.

Dataset	Recall@Top20%LOC	Effort@Top20%Recall	IFA	AUC	D2H	FAR
Primary dataset [42]	5/1/0	6/0/0	5/1/0	6/0/0	6/0/0	6/0/0
Additional dataset [10]	6/0/0	6/0/0	6/0/0	6/0/0	6/0/0	6/0/0